

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) Determine whether the three functions x , x^2 , and $1/x^2$ are linearly independent.

$$\begin{aligned}
 W(x, x^2, \frac{1}{x^2}) &= \begin{vmatrix} x & x^2 & 1/x^2 \\ 1 & 2x & -2/x^3 \\ 0 & 2 & 6/x^4 \end{vmatrix} \\
 &= x \begin{vmatrix} 2x & -2/x^3 \\ 2 & 6/x^4 \end{vmatrix} - 1 \cdot \begin{vmatrix} x^2 & 1/x^2 \\ 2 & 6/x^4 \end{vmatrix} \\
 &= x(12/x^3 + 4/x^3) - 1 \cdot (6/x^2 - 2/x^2) \\
 &= 12/x^2 > 0 \text{ if } x \neq 0 \\
 &\Rightarrow x, x^2, \frac{1}{x^2} \text{ are linearly independent on } (-\infty, 0) \cup (0, \infty)
 \end{aligned}$$

Problem 2. (6 points) Find the general solution for each of the following equations:

(a) $y''' - y'' - 4y' + 4y = 0$

$$\begin{aligned}
 r^3 - r^2 - 4r + 4 &= 0 \\
 r^2(r-1) - 4(r-1) &= 0 \\
 (r^2-4)(r-1) &= 0 \\
 r_1=2, r_2=-2, r_3=1 \\
 \Rightarrow y(t) &= C_1 e^{2t} + C_2 e^{-2t} + C_3 e^t
 \end{aligned}$$

(b) $y^{(4)} + 2y'' + y = 0$

$$\begin{aligned}
 r^4 + 2r^2 + 1 &= 0 \\
 (r^2 + 1)^2 &= 0 \\
 \Rightarrow r_1=r_2=i, r_3=r_4=-i \\
 \Rightarrow y(t) &= C_1 \cos t + C_2 t \cos t + C_3 \sin t + C_4 t \sin t
 \end{aligned}$$